

# Calvin-Khang Ta

COMPUTER VISION RESEARCHER

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## Education

### University of California, Riverside

Riverside, CA

PHD COMPUTER SCIENCE

2024

GPA: 3.5

### University of California, Riverside

Riverside, CA

B.S. COMPUTER SCIENCE

2019

GPA: 3.3

## Selected Publications

### DCVC-MB: Neural B-Frame Video Compression using State Space Models

Arjun\* Arora, Calvin-Khang\* Ta, Carlos M\*. Restrepo-Galeano, Kruthi Murali, Naga Akhil E S, Arunkumar Mohananchettiar, Jay N. Shingala, Tong Shao, Peng Yin, Sean McCarthy

*Under Review ICME 2026*

### Multi-modal Pose Diffuser: A Multimodal Generative Conditional Pose Prior

Calvin-Khang Ta, Arindam Dutta, Rohit Kundu, Rohit Lal, Hannah Dela Cruz, Dripta S. Raychaudhuri, Amit Roy-Chowdhury

*Under Review TPAMI*

### VOccl3D: A Video Benchmark Dataset for 3D Human Pose and Shape Estimation under real Occlusions

Yash Garg, Saketh Bachu, Arindam Dutta, Rohit Lal, Sarosij Bose, Calvin-Khang Ta, M Salman Asif, Amit Roy-Chowdhury

*Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025*

### Conformal Prediction and MLLM aided Uncertainty Quantification in Scene Graph Generation

Sayak Nag, Udit Ghosh, Calvin-Khang Ta, Sarosij Bose, Jiachen Li, Amit K Roy-Chowdhury

*Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR), 2025*

### POISE: Pose Guided Human Silhouette Extraction Under Occlusions

Arindam Dutta, Rohit Lal, Dripta S. Raychaudhuri, Calvin-Khang Ta, Amit K. Roy-Chowdhury

*Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024*

### Prior-guided Source-free Domain Adaptation for Human Pose Estimation

Dripta S. Raychaudhuri, Calvin-Khang Ta, Arindam Dutta, Rohit Lal, Amit K. Roy-Chowdhury

*Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2023*

### GAMA: Generative Adversarial Multi-Object Scene Attacks

Abhishek Aich\*, Calvin-Khang Ta\*, Akash Gupta, Chengyu Song, Srikanth V Krishnamurthy, M Salman Asif, Amit K Roy-Chowdhury

*Annual Conference on Neural Information Processing Systems, (NeurIPS), 2022*

### Poisson2Sparse: Self-supervised Poisson Denoising from a Single Image

Calvin-Khang Ta\*, Abhishek Aich\*, Akash Gupta, Amit K Roy-Chowdhury

*Medical Image Computing and Computer Assisted Intervention (MICCAI), 2022*

## Experience

### Dolby Laboratories

Sunnyvale, CA

SENIOR RESEARCHER

Oct. 2024 - PRESENT

- Spearheaded the R&D of next-generation end-to-end neural image/video compression (NVC) architectures, targeting SOTA performance against traditional codecs (e.g., VVC, AV1).
- Engineered novel conditioning techniques for variable-rate models, optimizing Rate-Distortion (R-D) trade-offs thus improving BD-rate by up to 2%.
- Secured intellectual property via patent disclosures on generative compression and authored manuscripts currently under review.
- Supervised PhD-level research on experimental architectures, delivering critical proof-of-concepts (PoC) that secured strategic IP and validated feasibility for future roadmaps.

- Conducting research into fundamental computer vision and machine learning problems.
- Developed automated image analysis pipelines for biological modeling research of Shoot Apical Meristem of Arabidopsis and increasing throughput of existing code base by 400%.
- Developed a novel self-supervised image enhancement algorithm through exploring sparse representations and implicit network priors which beat prior state of the art methods for Poisson denoising. Results published at MICCAI 2022.
- Developed a novel method for generating adversarial attacks on deep learning based multi object classifiers across domains leveraging joint image and language representations from pretrained CLIP models. Results published at NeurIPS 2022.
- Developed a novel multi-modal generative prior for enhancing the robustness of 3D body shape estimation leveraging images and textual descriptions. Currently under peer-review.
- Worked on tracking and object detection components for bio-metrics pipeline and developed pre-processing tools to prepare terabytes of video data for training and testing.

Dolby Laboratories

Sunnyvale, CA

VIDEO CODING RESEARCH INTERN

June 2023 - Sept. 2023

- Developed novel method for improving deep learning-based video compression models by exploiting redundancies in the trained entropy model.
- Proposed method improved the BD rate by 3% over internal baseline model.
- Optimized the code base and increased training speed by over 40% and reducing the amount of time needed training by 2 days.
- Work resulted in a patent disclosure and manuscript for peer-review is currently in progress.

Vimaan Robotics

Santa Clara, CA

COMPUTER VISION / MACHINE LEARNING RESEARCH INTERN

June 2022 - Sept. 2022

- Research and implemented state of the art image de-blurring algorithms for use on internal datasets and products.
- Investigate image quality metrics for quantifying the importance of samples in a dataset with respect to a given task.

Amazon.com

Various

SOFTWARE ENGINEERING INTERN

June. 2018, June. 2019, June. 2020

- Worked on a full-stack project under the Goodreads team as a part of a new UI/UX initiative.
- Responsible for converting existing monolith API endpoints to a scalable micro-service with AWS and implementing new modern front-end.
- Worked on the AWS Comprehend Team to develop an automated process for determining optimal hardware configuration.
- Utilized Bayesian optimization with a novel metric taking into account throughput and latency to estimate the optimal EC2 instance types.
- Worked with Seller Partner Promotions Team on the deal validation engine for Amazon.com
- Responsible for extending API support for batch and asynchronous processing.

Skills

|                                    |                                                                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Computer Vision   Machine Learning | Neural Networks, GANs, Object Detection, Tracking, 3D Pose Estimation, Image Restoration, Self-Supervised Learning, Adversarial Machine Learning, Vision-Language Models, Diffusion Models, Generative AI |
|                                    | Languages Python, C/C++, Java, MATLAB, Shell Script                                                                                                                                                       |
| Libraries Tools                    | Git, cmake, nmap, vagrant, ansible, Unity, OpenCV, numpy, pandas, Pytorch, AWS, Docker                                                                                                                    |

Academic Services

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|-------|-----------------------------------------------------------------------------------|
| 2024- | Reviewer, European Conference on Computer Vision                                  |
| 2024- | Reviewer, IEEE/CVF Conference on Computer Vision and Pattern Recognition          |
| 2023- | Reviewer, IEEE Transactions on Image Processing                                   |
| 2023- | Reviewer, IEEE Transactions on Pattern Analysis and Machine Intelligence          |
| 2024  | Reviewer, Artificial Intelligence with Biased or Scarce Data Workshop             |
| 2022  | Speaker, Google exploreCSR @ University of California Riverside                   |
| 2022  | Reviewer, IEEE CVPR Workshop on Fair, Data-efficient, and Trusted Computer Vision |